

Agricultural Analytics Platform

Overview

The Agricultural Analytics Platform is an end-to-end ELT data engineering project designed to consolidate agricultural datasets from multiple independent sources into a unified analytical data warehouse.

The project simulates a real-world scenario where different departments publish data independently. Production statistics, weather information, pesticide usage, and crop yield are stored in separate files with different structures, making analytical reporting difficult.

This platform automates the ingestion, validation, transformation, and integration of these datasets into a clean reporting model following the Medallion Architecture (Bronze → Silver → Gold).

Business Problem

Agricultural organizations often collect data from multiple agencies.

For example:

- Crop production data
- Rainfall measurements
- Average yearly temperature
- Pesticide usage
- Crop yield

Although each dataset is valuable individually, they cannot answer business questions when stored separately.

Questions such as:

- Does rainfall affect crop yield?
- Which countries produce the highest yield?
- Does pesticide usage improve production?
- Which crops perform better under different weather conditions?

require data integration before analysis.

Without a centralized pipeline, analysts must manually clean and combine datasets every time a report is needed.

This project automates that process.

Project Goals

- Build a production-style ELT pipeline.
 - Apply Medallion Architecture.
 - Standardize heterogeneous datasets.
 - Perform data quality validation.
 - Create analytical datasets optimized for reporting.
 - Produce dashboards and business insights.
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Source Datasets

Dataset	Purpose
Production	Agricultural production statistics
Rainfall	Average annual rainfall
Temperature	Average yearly temperature
Pesticides	Pesticide usage by country
Crop Yield	Final analytical measurements

Architecture

Raw Files



Bronze Layer



Data Validation



Silver Layer



Business Transformations

↓

Gold Layer

↓

Power BI Dashboard

Medallion Layers

Bronze

Purpose

Store raw source data exactly as received.

Characteristics

- No business transformations
 - Source preservation
 - Load metadata
 - Pipeline execution tracking
-

Silver

Purpose

Clean and standardize datasets.

Transformations include

- Standardizing column names
 - Removing duplicates
 - Handling missing values
 - Data type conversion
 - Data validation
 - Referential integrity checks
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Gold

Purpose

Produce business-ready datasets optimized for analytics.

The final dataset combines

- Production
- Rainfall
- Temperature
- Pesticides
- Crop Yield

into a single analytical model.

Technologies

Programming

- Python

Database

- PostgreSQL

Libraries

- Pandas
- SQLAlchemy
- psycopg2
- Pydantic

Architecture

- ELT
- Medallion Architecture

Version Control

- Git
- GitHub

Visualization (Upcoming)

- Power BI

Cloud (Future)

- Databricks
- Snowflake

Data Quality Checks

The pipeline validates

- Missing values
- Duplicate records
- Invalid data types
- Primary key uniqueness
- Join consistency
- Row count validation
- Pipeline execution logs

Business Questions Answered

Examples

- Which country has the highest crop yield?
- How does rainfall influence production?
- Does pesticide usage correlate with yield?
- Which crops perform consistently across years?
- Which regions show declining productivity?

Future Improvements

Version 2

- Incremental loading

Version 3

- Apache Airflow orchestration

Version 4

- Databricks implementation

Version 5

- Snowflake Data Warehouse

Version 6

- Power BI dashboards

Version 7

- Docker deployment

Version 8

- CI/CD pipeline

Repository Structure

```
agriculture-analytics-platform/
```

```
|  
├─ data/  
├─ docs/  
├─ sql/  
├─ python/  
├─ notebooks/  
├─ dashboards/  
├─ architecture/  
├─ tests/  
├─ images/  
└─ README.md
```

Learning Outcomes

This project demonstrates practical knowledge of

- Data Engineering
- ELT Pipeline Design
- Data Warehousing
- SQL
- Python
- Data Modeling
- Data Quality
- Logging
- Pipeline Monitoring
- Analytical Reporting

The project is intentionally built in multiple versions to simulate the evolution of a real production data platform rather than a one-time academic exercise.